

PRODUCT VISION DOCUMENT (PVD)

Project Title: ANUJNA: A Consent-Driven Unified Health Identity Platform

1. Vision Statement

To empower patients with complete visibility, control, and trust over their healthcare data through a secure, consent-driven digital health platform that enables authorized medical record sharing, transparent access governance, emergency healthcare support, and intelligent health data management.

2. Problem Statement

Healthcare records are often fragmented across hospitals, clinics, laboratories, and healthcare providers. Patients frequently carry physical documents, lose medical history, or struggle to share relevant records during consultations and emergencies.

Although digital health ecosystems are emerging, patients often lack meaningful understanding and control over who accesses their health information, for what purpose, and for how long.

Existing systems focus primarily on identity and record exchange, while challenges related to consent transparency, auditability, emergency access governance, and patient trust remain significant.

3. Product Vision

ANUJNA aims to become a patient-centric trust layer for healthcare data exchange.

Instead of merely storing medical records, the platform focuses on:

- Fine-grained consent management
- Transparent healthcare data sharing
- Patient-controlled authorization
- Emergency access governance
- Access auditability
- Intelligent health record organization

The system ensures that healthcare information remains accessible when needed while preserving patient autonomy, privacy, and accountability.

4. Target Users

i) Primary Users

1. Patients

Individuals managing and sharing their healthcare records.

2. Doctors

Healthcare professionals requesting access to patient records for diagnosis and treatment.

3. Hospitals and Clinics

Healthcare institutions managing authorized access workflows.

ii) Secondary Users

4. Laboratory Personnel

Uploading diagnostic reports and test results.

5. Healthcare Administrators

Monitoring compliance, access requests, and operational workflows.

6. Emergency Care Providers

Accessing critical records during emergency situations under governed workflows.

5. Core Value Proposition

ANUJNA provides:

- Unified health identity
- Consent-driven access control
- Fine-grained authorization
- Complete access transparency
- Emergency healthcare support
- Intelligent health record organization

Patients remain the primary owners and decision-makers for their healthcare information.

6. Product Differentiators

D1. Fine-Grained Consent Architecture

Patients can define:

- Who can access records
- Which records can be accessed

- Purpose of access
- Access duration
- Access scope

Example:

Doctor A may access cardiology records for 48 hours for consultation purposes only.

D2. Time-Bound Consent

Access permissions automatically expire after approved durations.

D3. Consent Revocation

Patients may revoke previously granted permissions at any time.

D4. Break-the-Glass Emergency Access

In emergency situations, authorized healthcare providers may request temporary emergency access.

All emergency actions are:

- Logged
- Audited
- Traceable
- Patient-notified

D5. Transparent Audit Dashboard

Patients can view:

- Who accessed records
- When access occurred
- What information was viewed
- Why access was granted

D6. AI-Assisted Health Timeline

Medical history is automatically organized into a chronological timeline for easier understanding and clinical review.

7. MVP Scope

Identity Management

- Registration
- Authentication
- Patient profile

Health Record Management

- Upload reports
- Organize records
- Record categorization

Consent Management

- Grant access
- Revoke access
- Time-bound permissions

Doctor Access Workflow

- Access request
- Access approval
- Authorized record viewing

Emergency Access Workflow

- Emergency request
- Temporary authorization
- Audit logging

Auditability

- Access logs
- Activity history
- Transparency dashboard

8. Phase 2 Scope

- OCR-based report extraction
- AI-generated health summaries
- Medical terminology simplification
- Consent templates
- Advanced analytics
- Suspicious access detection
- Email and notification services

9. Future Scope

- FHIR interoperability
- National healthcare ecosystem integration
- Wearable device integration
- Advanced privacy-preserving architectures
- Consent risk analysis engine
- Healthcare research insights
- Federated healthcare exchange networks

10. Success Criteria

The platform will be considered successful if it can:

- Demonstrate patient-controlled healthcare data sharing
- Implement fine-grained consent management
- Support emergency healthcare workflows
- Maintain complete auditability
- Provide transparent access visibility
- Serve as a scalable foundation for future healthcare interoperability

11. Proposed Technology Stack

1. **Frontend:** Next.js, TypeScript and Tailwind CSS
2. **Backend:** Next.js APIs
3. **Database:** PostgreSQL
4. **ORM:** Prisma
5. **Authentication:** JWT and OAuth
6. **Deployment:** Vercel
7. **Version Control:** GitHub

12. Development Methodology

Recommended Model: **Agile + Spiral Hybrid**

Reason:

- Supports iterative development
- Encourages risk-driven design decisions
- Suitable for evolving healthcare requirements
- Allows continuous refinement of consent architecture and security mechanisms

Version: **PVD v1.0**

Status: **Approved** for Requirements Engineering Phase